

free and open source software? Well, here is the justification. GPT-1 and GPT-2 are fully open source (licensed under the MIT License) with their training code and model weights accessible on GitHub. It's only from GPT-3 that OpenAI (the organisation that developed ChatGPT) shifted to keeping their models proprietary.

That said, OpenAI has recently released two new open-weight models—gpt-oss-120b and gpt-oss-20b—licensed under the Apache License 2.0. These are the first open-weight releases from OpenAI since GPT-2 back in 2019. Unlike fully open source models, 'open-weight' models provide access to the trained parameters (weights) of the model, enabling developers to inspect, fine-tune, and deploy them locally or on their own infrastructure without requiring the original training data. This significant move looks like a renewed commitment from OpenAI towards openness, making it highly likely we will see more open source releases in the future. This makes the topic especially relevant and valuable for the OSFY readership.

In this article, we will take a closer look at a selection of custom GPTs—both the surprisingly useful and the delightfully quirky. Whether you are looking to boost productivity or simply have some fun, there is likely a custom GPT that fits the bill. To give you a glimpse of what these custom GPTs can do, I will be sharing responses generated by the following selected ones.

Let us begin on a lighter note by exploring a custom GPT called Monday, which comes with the intriguing tagline: 'A personality experiment. You may not like it. It may not like you'. Monday is part of OpenAI's ongoing exploration into stylized, personality-driven custom GPTs—models designed to exhibit distinct voices, often witty, sarcastic, or blunt. To test it out, I asked Monday about my research—something I take great pride in. The response I received,

laced with sarcasm (or perhaps brutal honesty!), was so unexpected that it made me momentarily question my life choices.

"Ah yes. You took a theorem from the 1970s—probably scribbled on a napkin by two sleep-deprived mathematicians—and said, "Hey, what if I improved this by 0.0001?" And then you did. And then you published it. And now you're a legend in the three-person subreddit that follows oriented graph diameters."

Yes, Monday turned out to be surprisingly pessimistic—and, at times, downright irritating.

To boost my morale, I turned to Scholar GPT, a custom GPT designed for academic support and research exploration. I asked Scholar GPT to explain oriented diameter of a graph, one of my research areas, which has applications in operations research. The response was clear, insightful, and reaffirmed the value of my work—exactly the kind of encouragement I needed after my encounter with Monday. An example response generated by Scholar GPT is shown below.

"In graph theory, the oriented diameter of an undirected connected graph refers to the smallest possible diameter among all its strongly connected orientations. That is, one assigns a direction to each edge such that the resulting directed graph remains strongly connected, and the goal is to minimise the maximum distance between any two vertices."

But later, I began to wonder if I was just going through a streak of bad luck—asking a sarcastic AI tool like Monday to evaluate my research, getting annoyed by its response, and then turning to another AI tool just to restore my self-esteem. Not exactly the healthiest way to use AI, I thought. Maybe it was time to consult an astrologer and make sense of it all. And to my surprise, there is a custom GPT for that too—Astrology Birth Chart GPT

(not ChatGPT but Chart GPT). What luck! I entered my date, time, and place of birth, and instantly received detailed insights into my supposed past karma, current dilemmas, and future missteps. Since it was all praise, I decided not to include a sample paragraph generated by Astrology Birth Chart GPT. But now I am a bit worried—given that I was logged into ChatGPT, was it really making astrological predictions, or was it just cleverly quoting from my online CV? Maybe it is time I consult a real astrologer after all.

The jokes are over—now it is time to get serious. In the rest of this article, we will focus on two key topics. First, we will examine how these various custom GPTs differ from ChatGPT. Are they truly distinct models, or is it just ChatGPT operating behind the scenes under different names? Once we answer this crucial question, we will explore a curated list of custom GPTs that are indispensable for boosting the productivity of professionals, researchers, and students alike.

Now, let us explore how these hundreds of custom GPTs may potentially offer something beyond what ChatGPT provides. I want to emphasise the word 'potentially'. In theory, a custom GPT could deliver additional features or services that go beyond ChatGPT's capabilities, but in practice, this may not always be the case. So, it is wise to approach these so-called next-generation custom GPTs with a healthy dose of caution.

Almost all custom GPTs available today are built on top of a foundation LLM (large language model), often from OpenAI (such as GPT-3 or GPT-4) or from other similar providers. At this foundational stage, the custom GPT is only as capable as the underlying model it uses.

The next step is specialisation, which is generally achieved in one of two ways: fine-tuning or prompt engineering. In fine-tuning, the base GPT model is further trained on